

EAS 6134:

Inverse Methods in EAS

Nonlinear inversion

18 March 2026

(several slides borrowed from C. and S. Constable)

To invert non-linear forward problems we often linearize around a starting model:

$$\hat{\mathbf{d}} = f(\mathbf{m}_1) = f(\mathbf{m}_0 + \Delta\mathbf{m}) \approx f(\mathbf{m}_0) + \mathbf{J}\Delta\mathbf{m}$$

using a matrix of derivatives

$$J_{ij} = \frac{\partial f(x_i, \mathbf{m}_0)}{\partial m_j}$$

and a model perturbation

$$\Delta\mathbf{m} = \mathbf{m}_1 - \mathbf{m}_0 = (\delta m_1, \delta m_2, \dots, \delta m_N)$$

Now our expression for χ^2

$$\chi^2 = ||\mathbf{W}\mathbf{d} - \mathbf{W}\hat{\mathbf{d}}||^2$$

Is then

$$\chi^2 \approx ||\mathbf{W}\mathbf{d} - \mathbf{W}f(\mathbf{m}_0) + \mathbf{WJ}\Delta\mathbf{m}||^2$$

For a least squares solution we solve in the usual way by differentiating and setting to zero to get a linear system:

$$\beta = \alpha \Delta \mathbf{m}$$

where

$$\beta = (\mathbf{WJ})^T \mathbf{W}(\mathbf{d} - f(\mathbf{m}_0))$$

$$\alpha = (\mathbf{WJ})^T \mathbf{WJ} \quad .$$

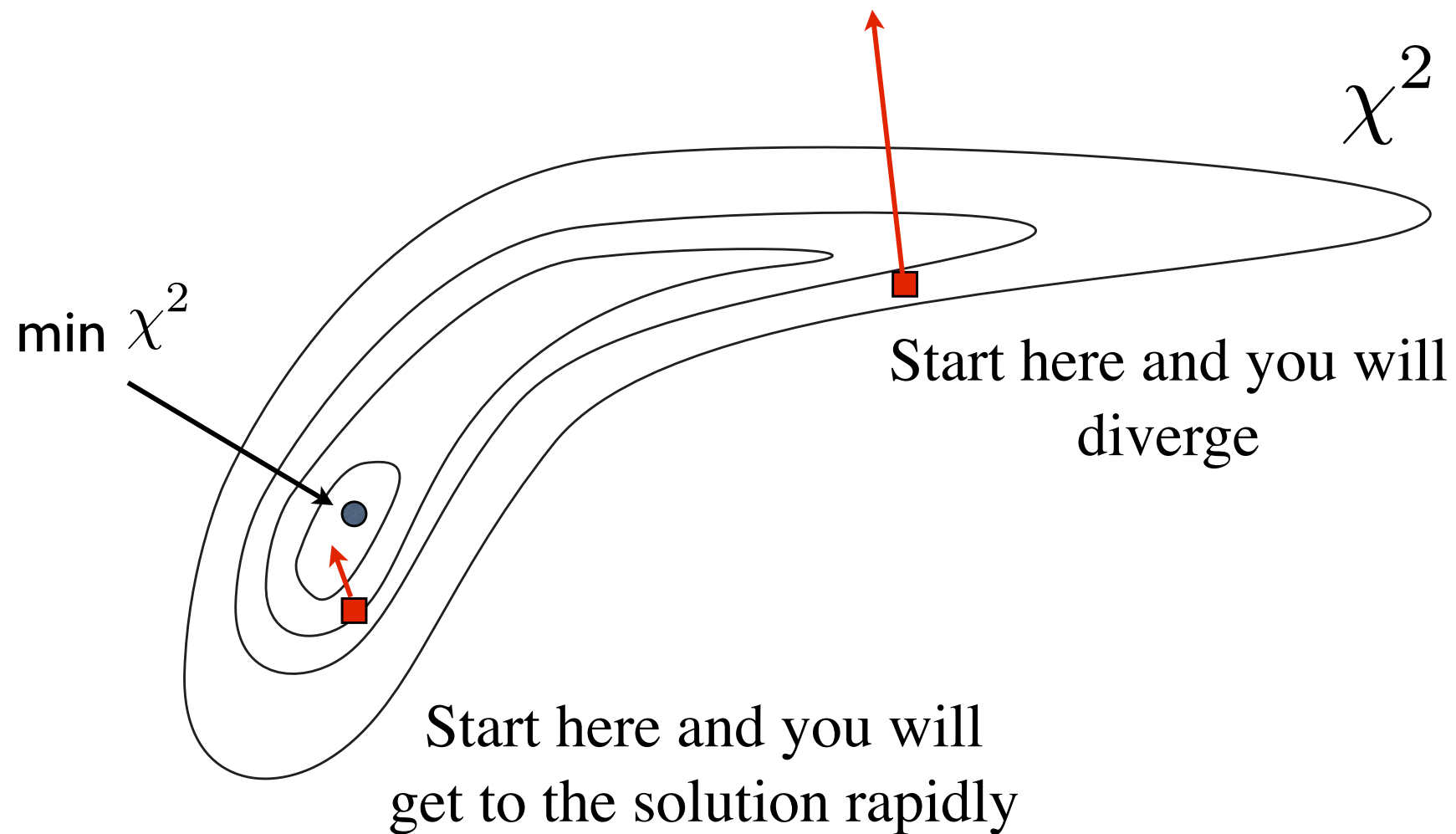
So, given a starting model $\mathbf{m}_0$ we can find an update $\Delta \mathbf{m}$ and iterate until we converge. (This is Gauss-Newton.)

$$\Delta \mathbf{m} = \alpha^{-1} \beta$$

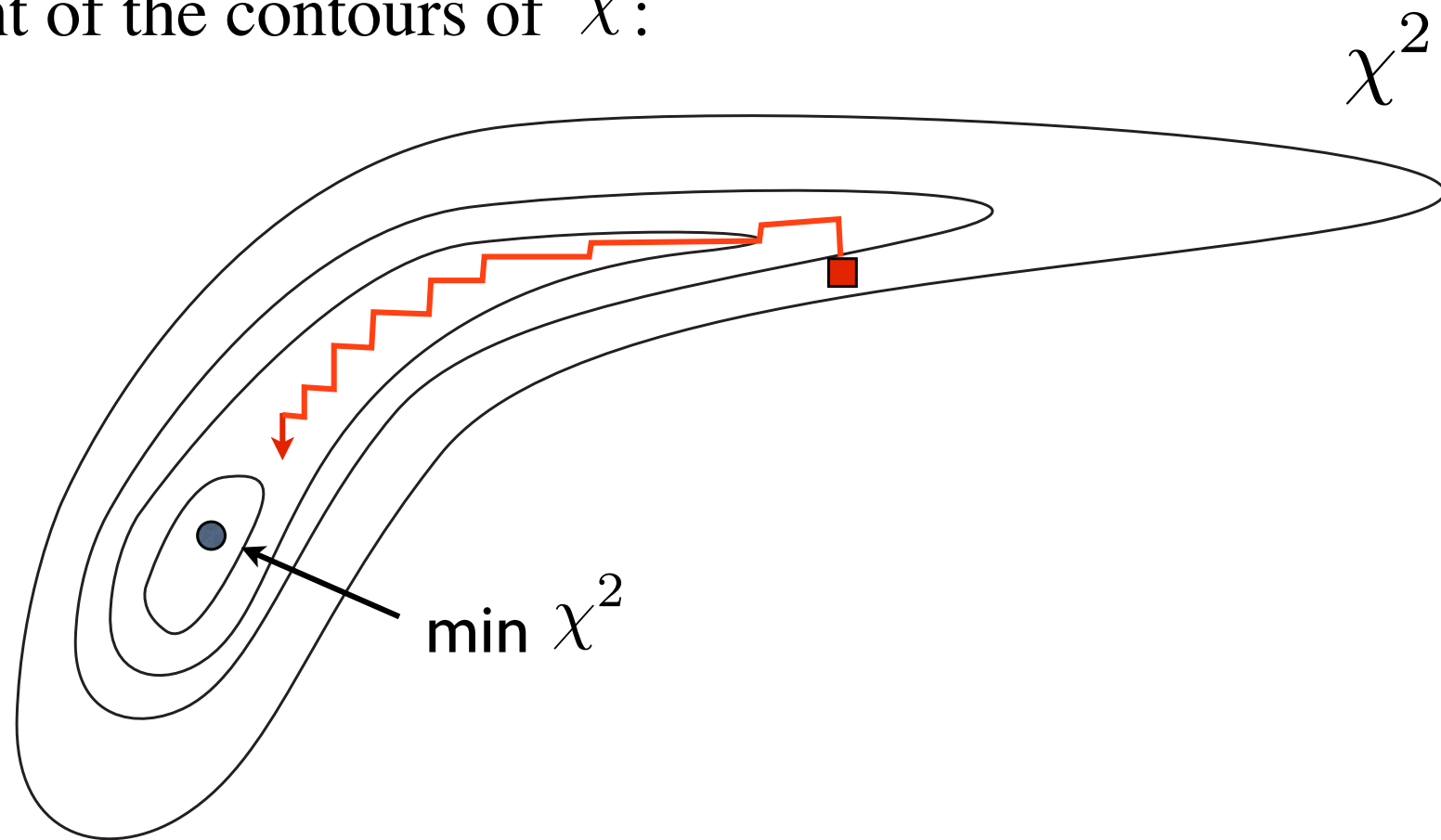
This can only work for small N (it isn't even defined for $N > M$). Even for very sparse parameterizations, it rarely works without modification.

* N here means number of model parameters

You need to start “linearly close” to the solution for this to work. Long, thin, valleys in χ^2 space are common and present problems for Gauss-Newton methods.



Another approach is “steepest descent”, in which you go down the gradient of the contours of χ^2 :



These solutions are of the form

$$\Delta \mathbf{m} = \mu (\mathbf{WJ})^T [\mathbf{W}(\mathbf{d} - f(\mathbf{m}_0))]$$

but the steps are always orthogonal and the step size is proportional to the slope, so this method stalls near the solution

The **Marquardt** method combines the **steepest descent** method and **Newton** method in one algorithm by modifying the curvature matrix:

$$\begin{aligned}\alpha_{jk} &= \alpha_{jk}(1 + \lambda) && \text{for } j = k \\ \alpha_{jk} &= \alpha_{jk} && \text{for } j \neq k \quad .\end{aligned}$$

When λ is large, this reduces to steepest descent. When λ is small, it reduces to the Newton method. Starting with large λ and then reducing it close to the solution works very well.

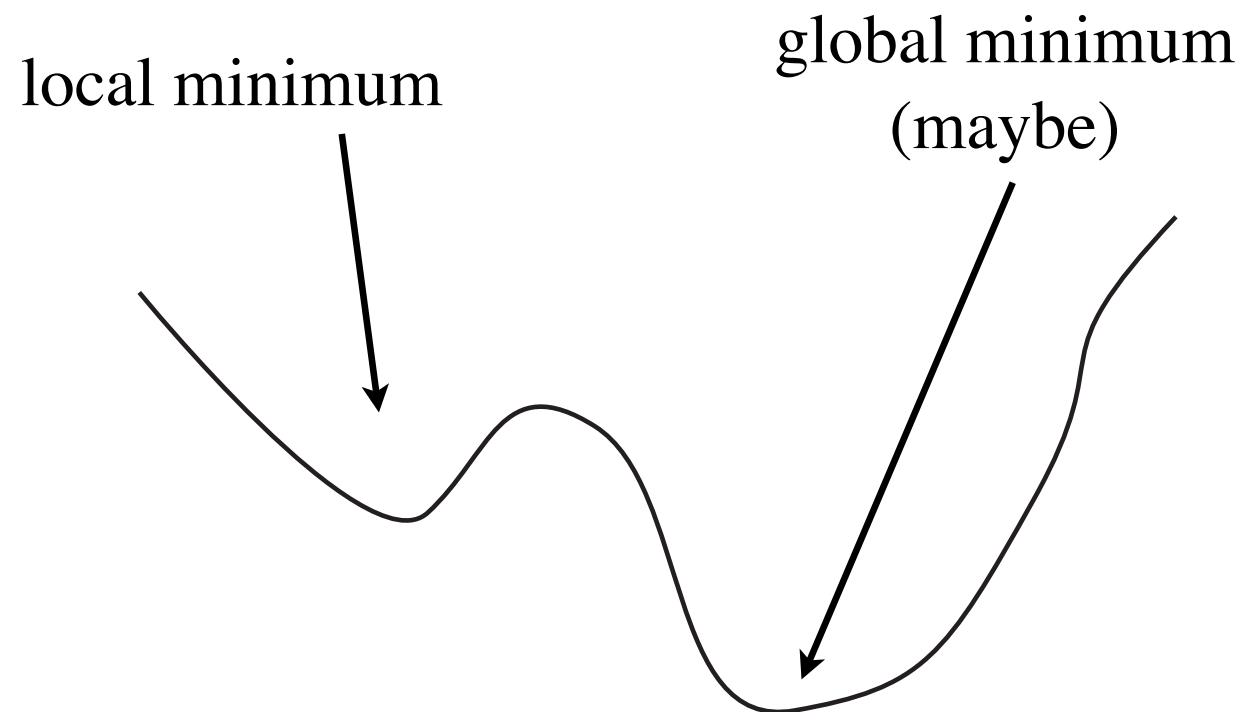
For problems that are naturally discretely parameterized, Marquardt is hard to beat. For sparse parameterizations of infinite dimensional models, the parameterization (e.g. number of layers chosen) has a big influence on the outcome.

Global versus local minima:

For nonlinear problems, there are no guarantees that Gauss-Newton will converge.

There are no guarantees that if it does converge the solution is a global one.

The solution might well depend on the starting model.



If you increase N too much, even with the Marquardt approach the solution goes unstable.

If N is big then the solutions become unstable, oscillatory, and generally useless

Almost all inversion today incorporates some type of regularization, which minimizes some aspect of the model as well as fit to data:

$$U = ||\mathbf{R}\mathbf{m}_1||^2 + \mu^{-1} (||\mathbf{W}\mathbf{d} - \mathbf{W}f(\mathbf{m}_1)||^2)$$

where $\mathbf{R}\mathbf{m}$ is some measure of the model and μ is a trade-off parameter or Lagrange multiplier.

$$U = ||\mathbf{R}\mathbf{m}_1||^2 + \mu^{-1} (||\mathbf{W}\mathbf{d} - \mathbf{W}f(\mathbf{m}_1)||^2)$$

In 1D a typical $\mathbf{R}$ might be:

$$\mathbf{R}_1 = \begin{pmatrix} -1 & 1 & 0 & 0 & 0 & \dots & 0 \\ 0 & -1 & 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & -1 & 1 & 0 & \dots & 0 \\ & & \ddots & & & \ddots & \\ & & & & -1 & 1 \end{pmatrix}$$

m ₁	-1	
m ₂	+1	-1
m ₃		+1 -1
m ₄		+1 -1
m ₅		+1 -1
m ₆		+1 -1
m ₇		+1 -1
m ₈		+1

which extracts a measure of slope. This stabilizes the inversion, creates a unique solution, and manufactures models with useful properties.

This is easily extended to 2D and 3D modelling.

$$U = ||\mathbf{R}\mathbf{m}_1||^2 + \mu^{-1} (||\mathbf{W}\mathbf{d} - \mathbf{W}f(\mathbf{m}_1)||^2)$$

When μ is small, model roughness is ignored and we try to fit the data. When μ is large, we smooth the model at the expense of data fit.

One approach is to choose μ and minimize U by least squares. There are various sets of machinery to do this (Newton, quasi-Newton, conjugate gradients, etc.). With many of these methods μ must be chosen by trial and error, increasing the computational burden and introducing some subjectivity.

Picking μ *a priori* is simply choosing how rough your model is compared to the data misfit. But, we've no idea how rough our model should be. However, we ought to have a decent idea of how well our data can be fit.

The Occam approach is to introduce some acceptable fit to the data (χ_*^2) and minimize:

$$U = ||\mathbf{R}\mathbf{m}_1||^2 + \mu^{-1} (||\mathbf{W}\mathbf{d} - \mathbf{W}f(\mathbf{m}_1)||^2 - \chi_*^2)$$

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Linearizing:

$$U = ||\mathbf{R}\mathbf{m}_1||^2 + \mu^{-1} (||\mathbf{W}\mathbf{d} - \mathbf{W}(f(\mathbf{m}_0) + \mathbf{J}(\mathbf{m}_1 - \mathbf{m}_0))||^2 - \chi_*^2)$$

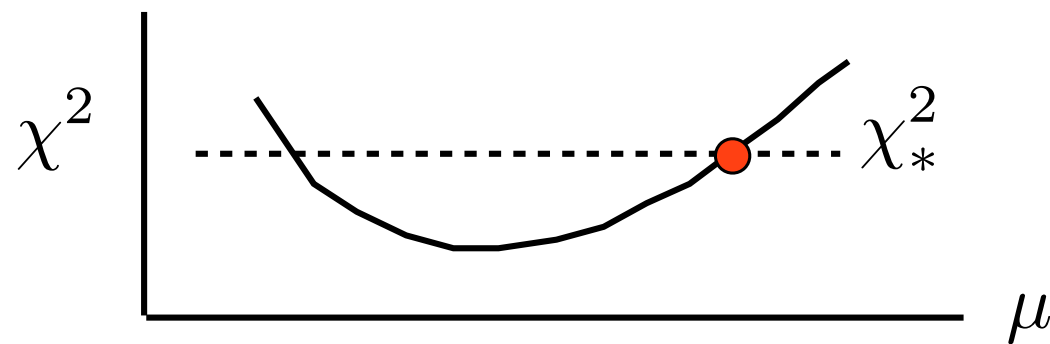
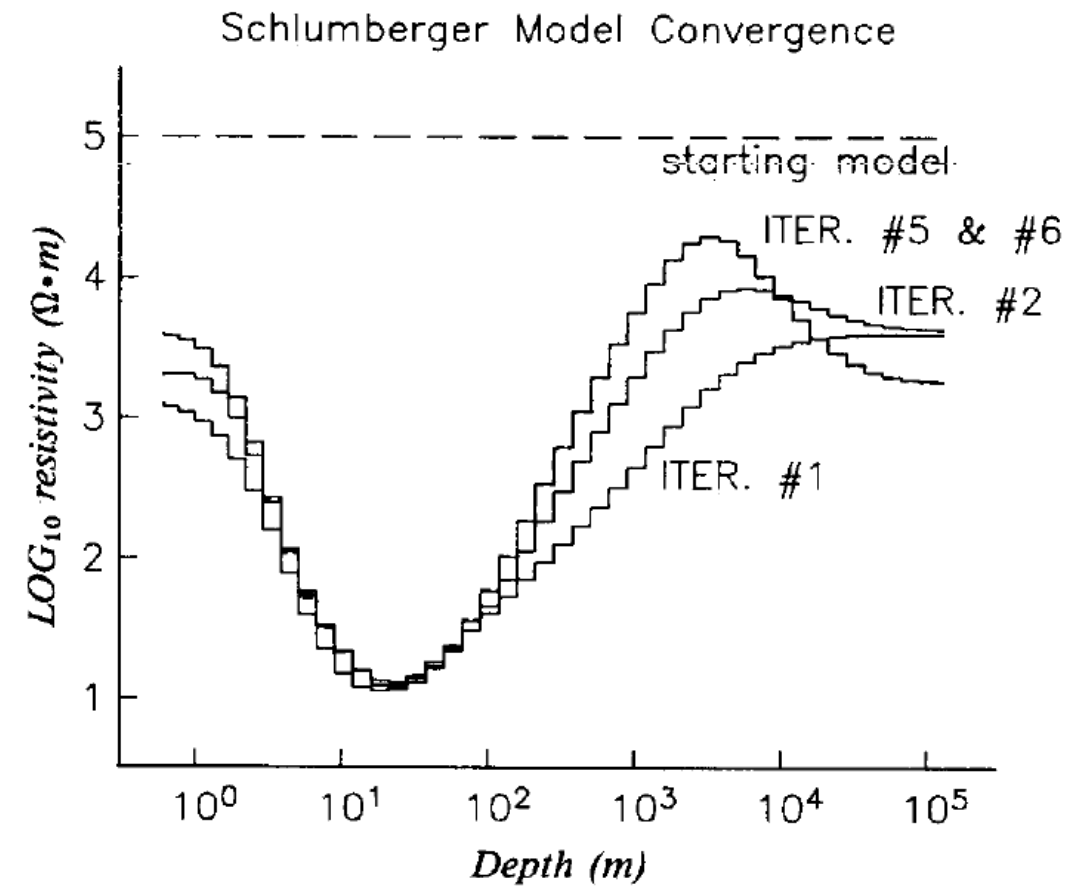
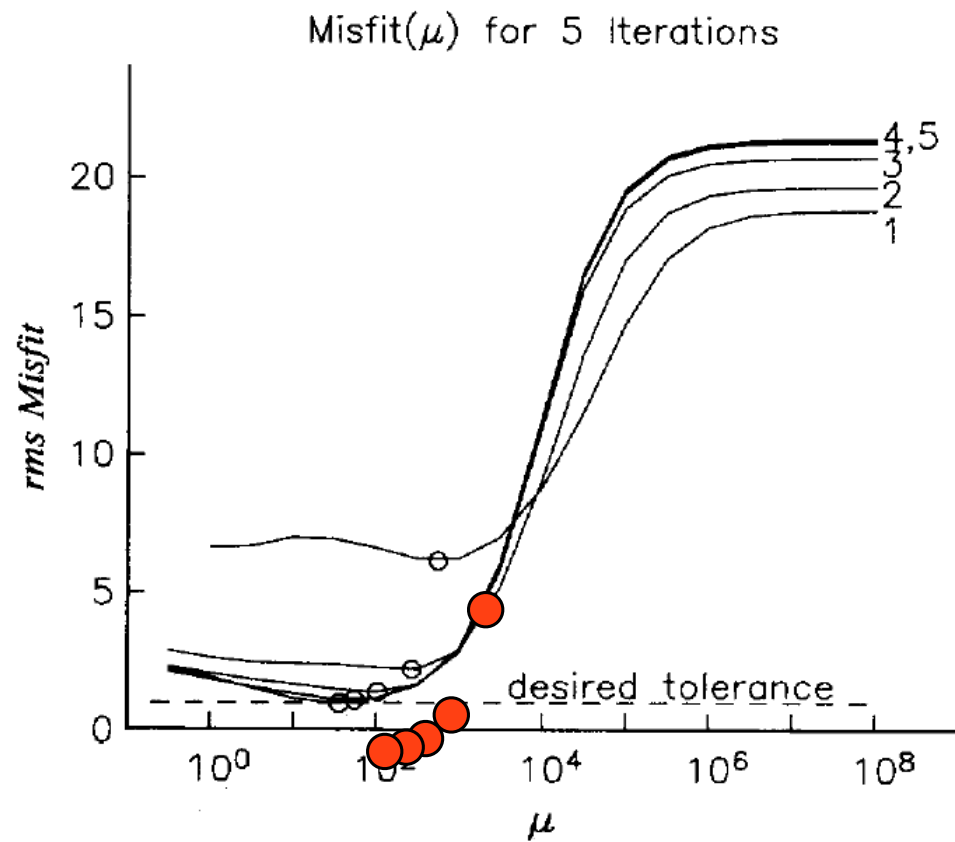
After differentiation and setting to zero we get

$$\mathbf{m}_1 = [\mu\mathbf{R}^T\mathbf{R} + (\mathbf{W}\mathbf{J})^T\mathbf{W}\mathbf{J}]^{-1}(\mathbf{W}\mathbf{J})^T\mathbf{W}(\mathbf{d} - f(\mathbf{m}_0) + \mathbf{J}\mathbf{m}_0) \quad .$$

The only thing we need is to find the right value for μ .

(Note we are solving for the next model $\mathbf{m}_1$ directly instead of $\Delta\mathbf{m}$.
Bob Parker calls these “leaping” and “creeping” algorithms.)

Occam finds μ by carrying out a line search to find the ideal value. Before χ_*^2 is reached, we minimize χ^2 . After χ_*^2 is reached we choose the μ which gives us exactly χ_*^2

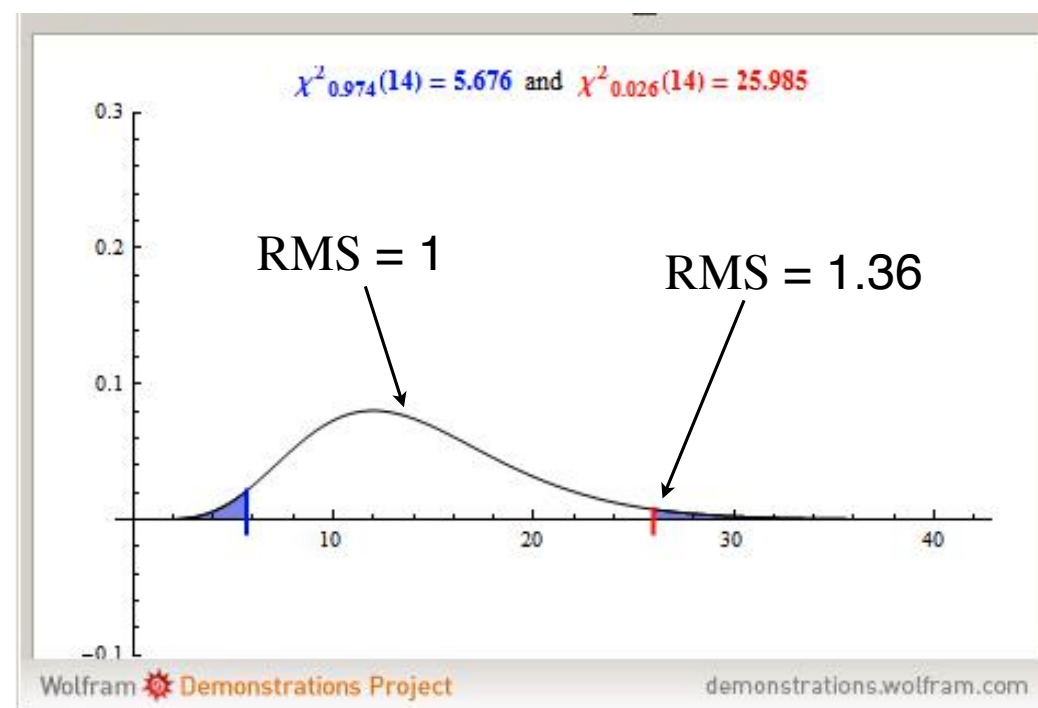


How to choose χ_*^2 ?

For zero-mean, Gaussian, independent errors, the sum-square misfit

$$\chi^2 = \|\mathbf{W}\mathbf{d} - \mathbf{W}\hat{\mathbf{d}}\|^2$$

is chi-squared distributed with M degrees of freedom. The expectation value is just M , which corresponds to an *RMS* of one, and so this could be a reasonable target misfit. Or, one could look up the 95% (or other) confidence interval for chi-squared M .



So, if our errors are well estimated and well behaved, this provides a statistical guideline for choosing χ^2_* .

Errors come from

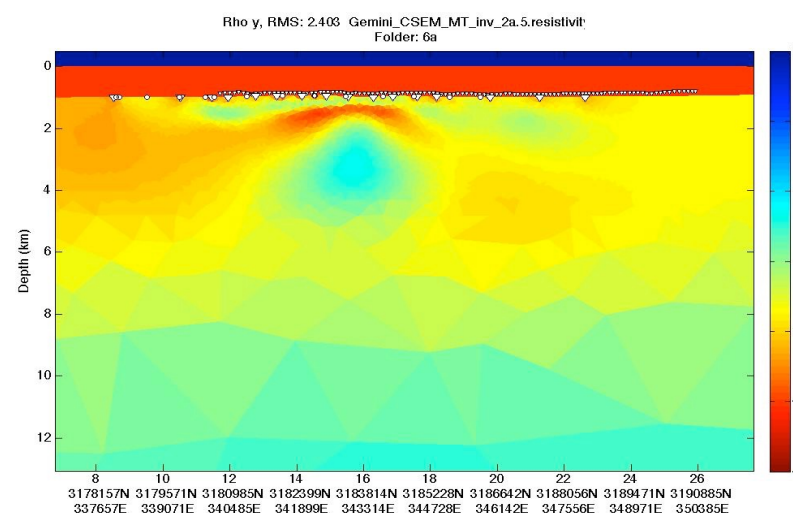
- statistical processing errors
- systematic errors such as instrument calibrations, and
- “geological or geophysical noise” (our inability to parameterize fine details of geology or extraneous physical processes).

Instrument noise should be captured by processing errors, but some error models assume stationarity (i.e. noise statistics don't vary with time). In practice, we may well only have a good handle on processing errors - everything else is lumped into a noise floor.

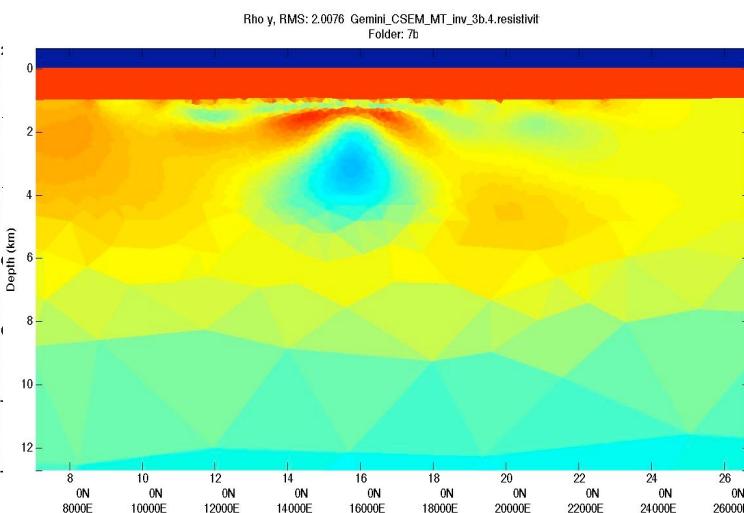
Note that it is meaningless to think in terms of $M-N$ degrees of freedom when N is not only very large, but also contains huge amounts of correlation from the regularization.

Even with well-estimated errors, choice of misfit can still be somewhat subjective.

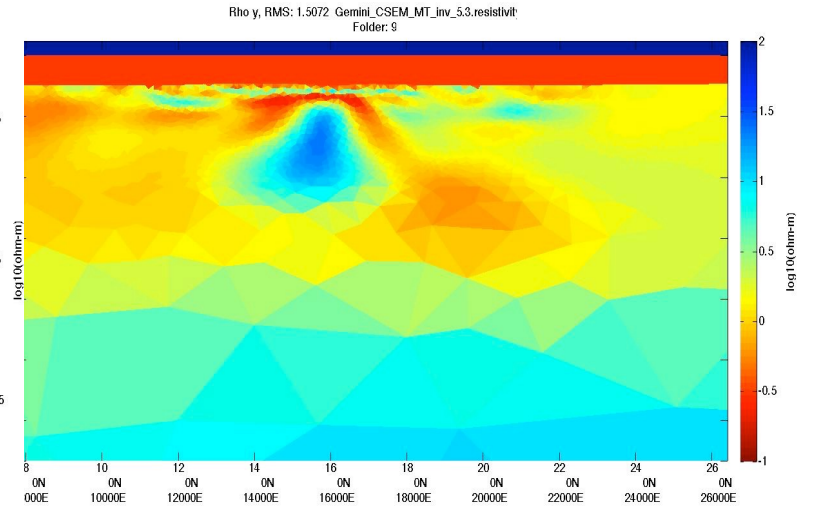
Joint 2D inversion of marine CSEM (3 frequencies, no phase) and MT (Gemini salt prospect, Gulf of Mexico):



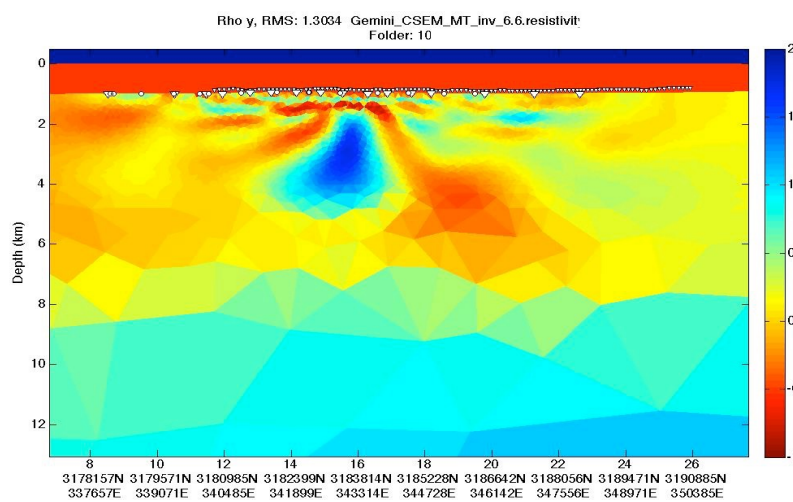
Target misfit 2.4



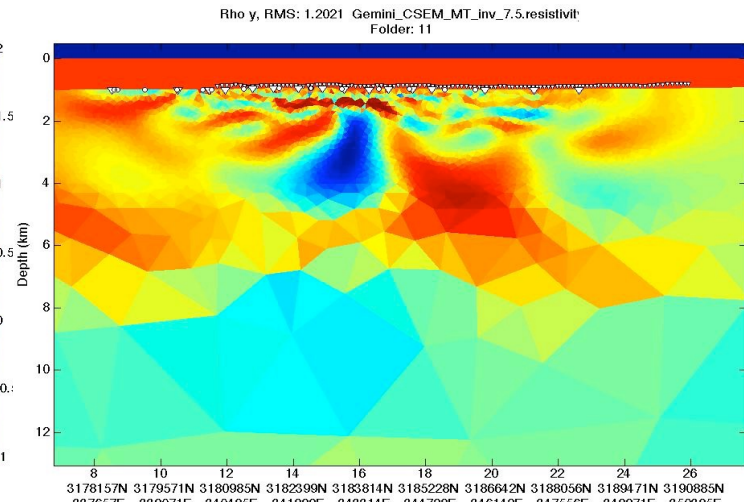
Target misfit 2.0



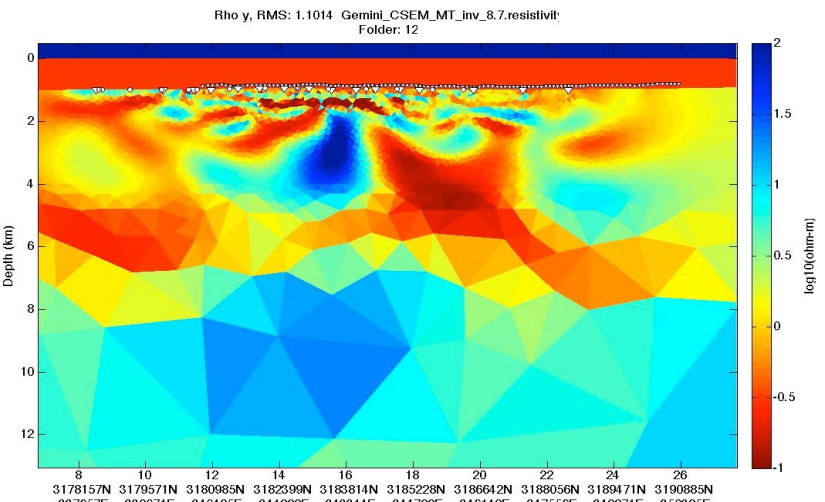
Target misfit 1.5



Target misfit 1.3

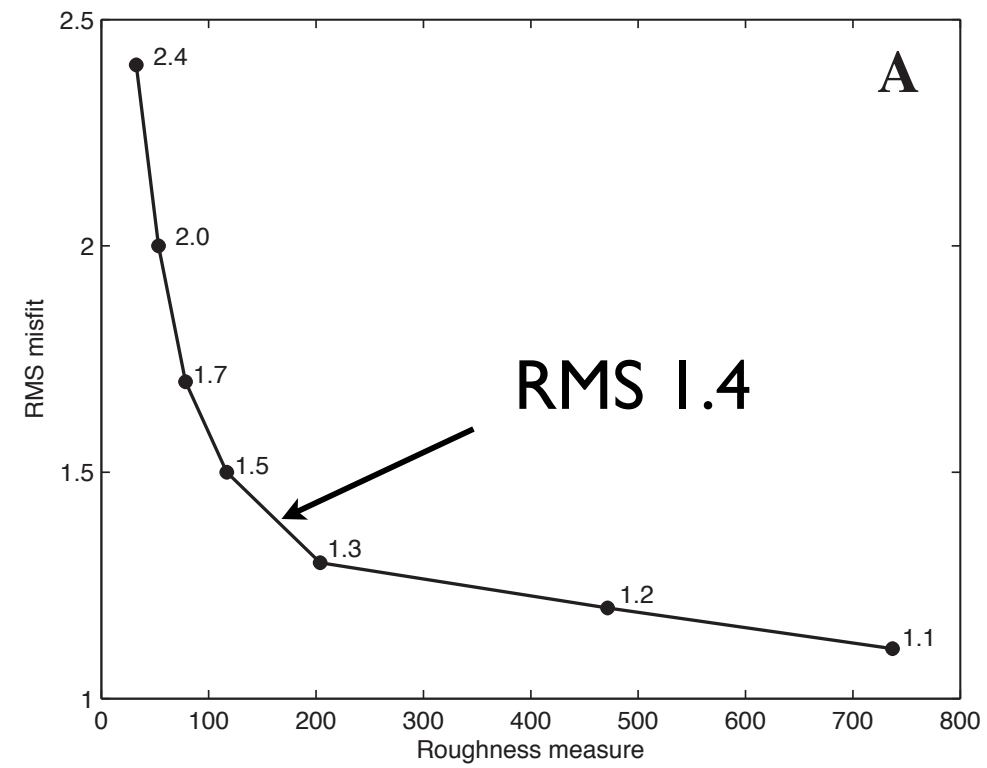


Target misfit 1.2

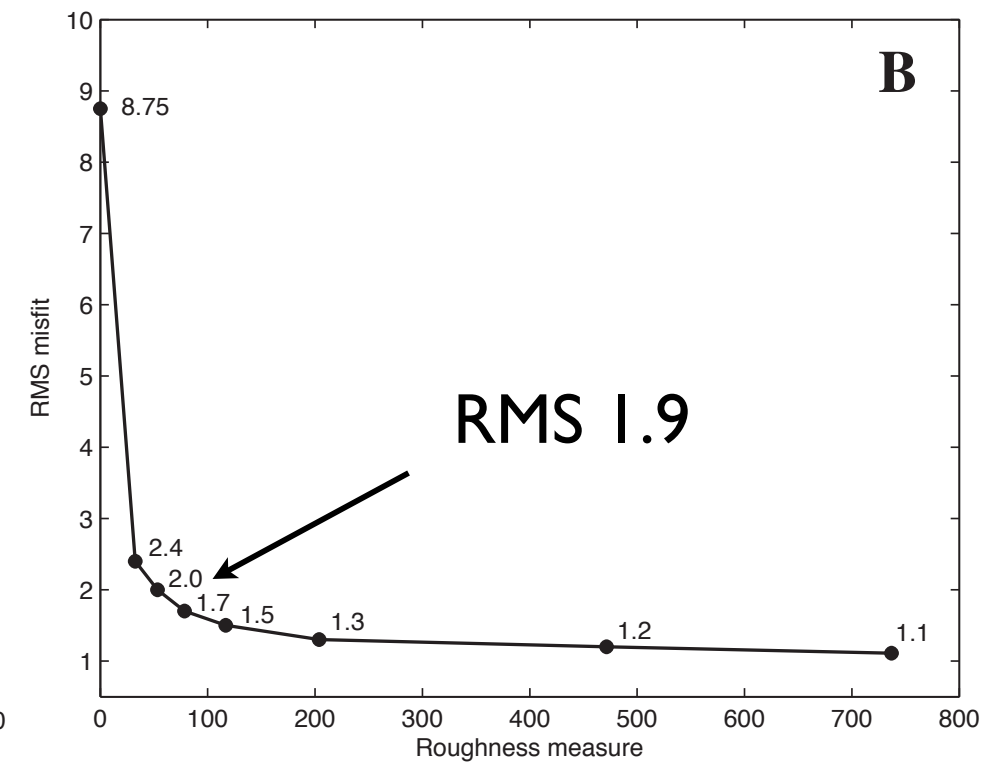
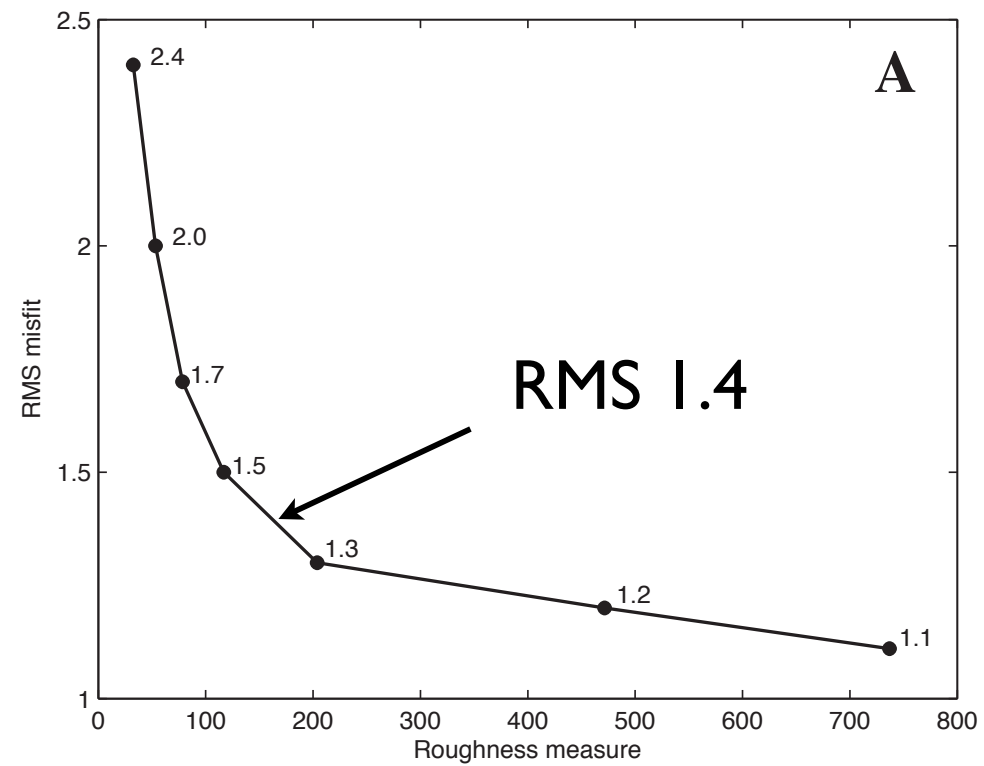


Target misfit 1.1

Beware of trade-off (“L”) curves:



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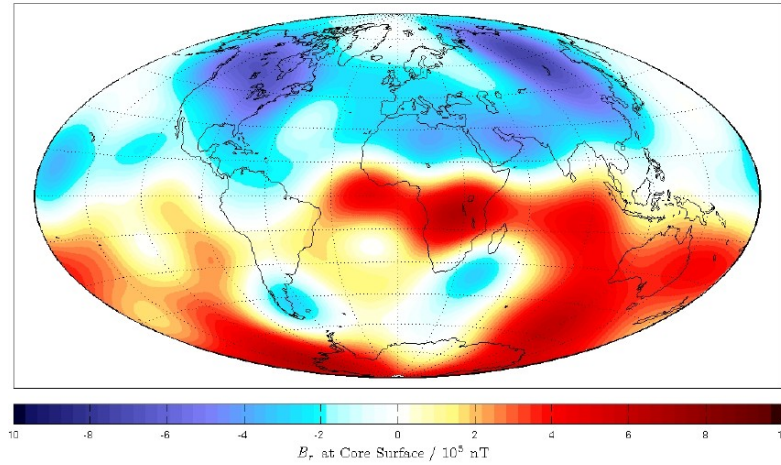
(they are not as objective as proponents say...)

Choice of Misfit Again

Regularization

Solution on the trade-off curve

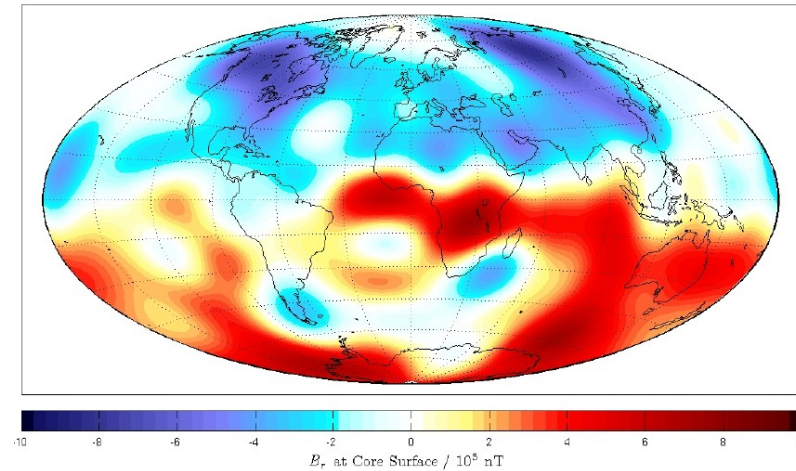
Damping parameter $\lambda = 1 \cdot 10^{-7} [\text{nT}]^{-2}$



Regularization

Solution on the trade-off curve

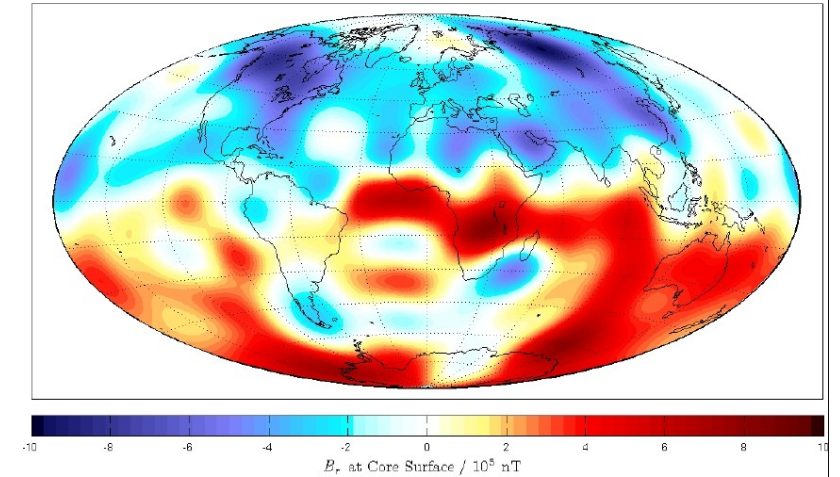
Damping parameter $\lambda = 1 \cdot 10^{-8} [\text{nT}]^{-2}$



Regularization

Solution on the trade-off curve

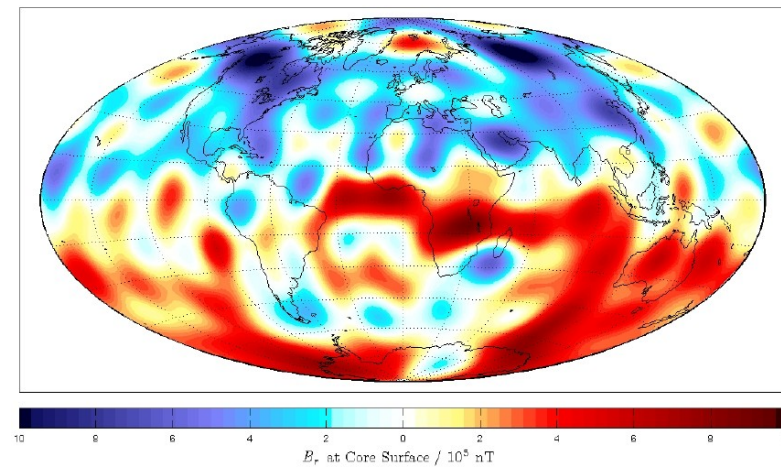
Damping parameter $\lambda = 1 \cdot 10^{-9} [\text{nT}]^{-2}$



Regularization

Solution on the trade-off curve

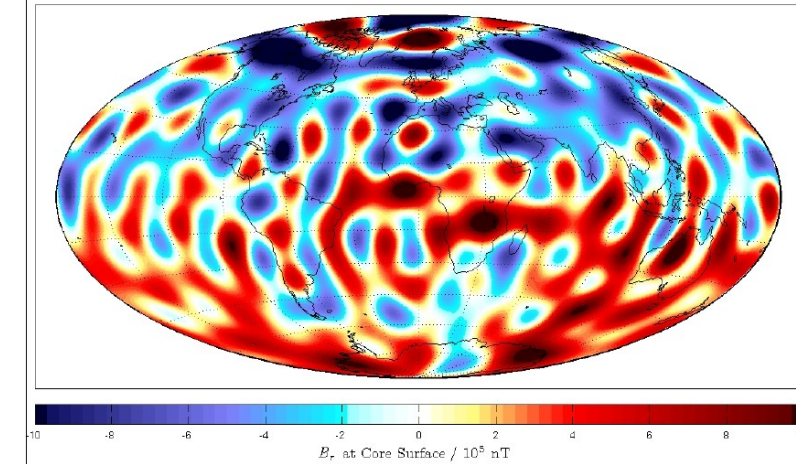
Damping parameter $\lambda = 1 \cdot 10^{-10} [\text{nT}]^{-2}$



Regularization

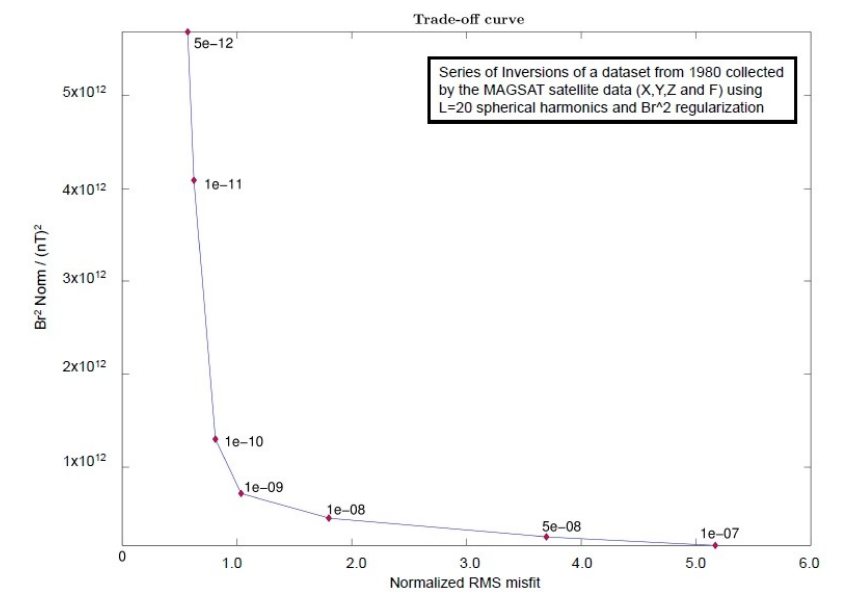
Solution on the trade-off curve

Damping parameter $\lambda = 1 \cdot 10^{-11} [\text{nT}]^{-2}$



Regularization

Example of core surface field trade-off curve



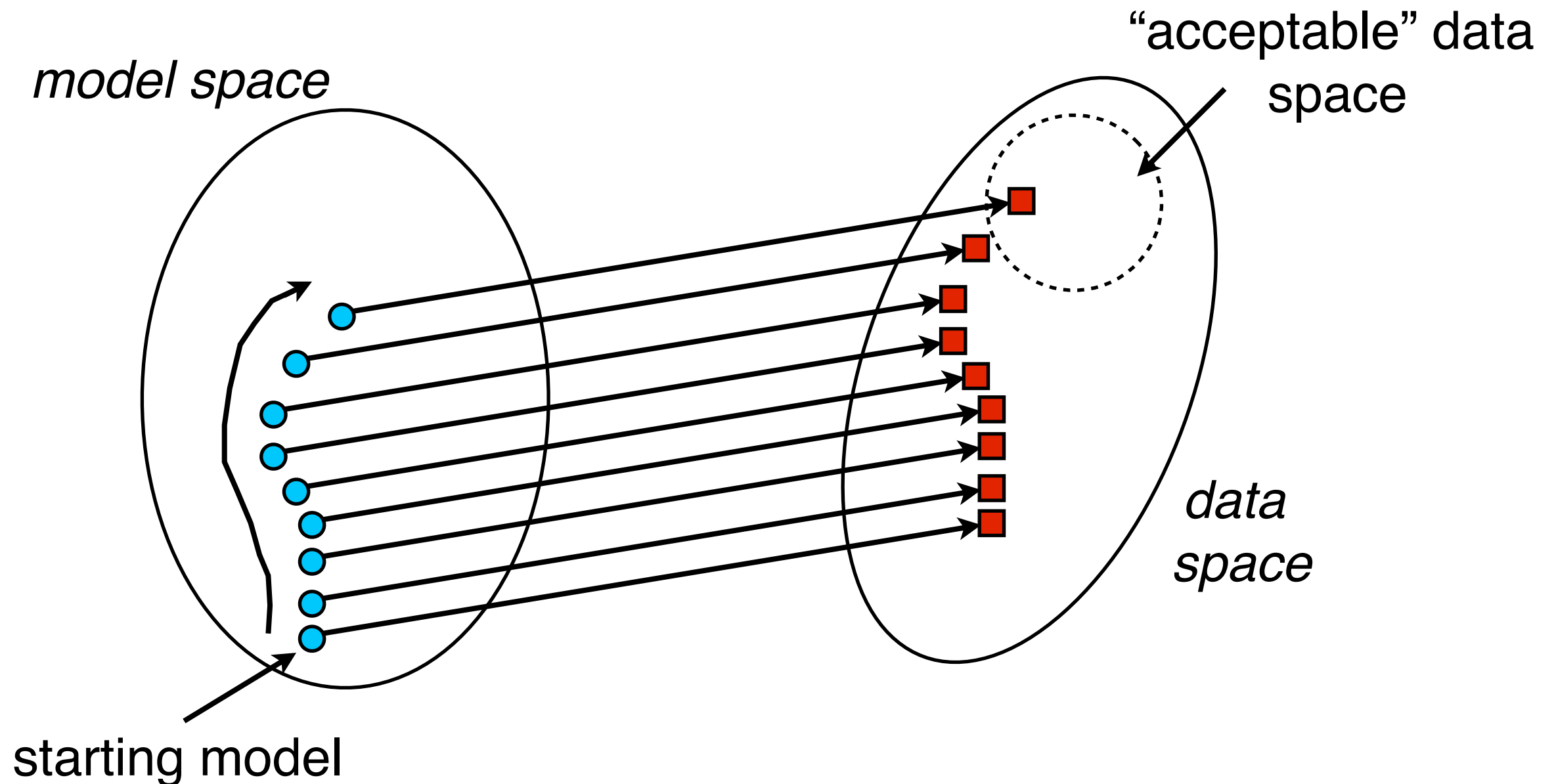
So far we have covered ‘Deterministic’ inversion methods

Deterministic

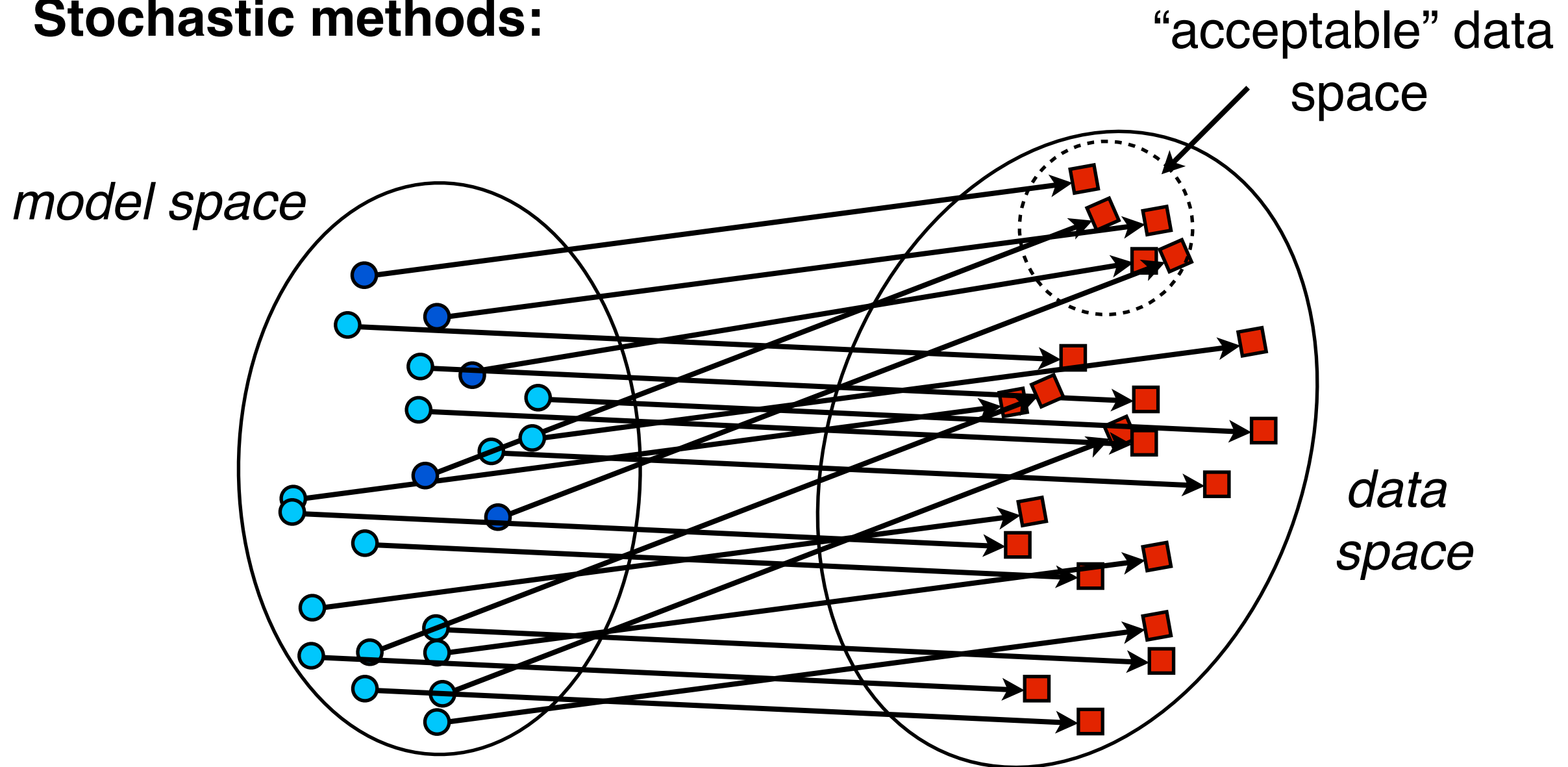
Newton Algorithms

Steepest descent

Conjugate Gradients



Stochastic methods:



A useful approach, largely restricted to simple problems (because millions of models required), with most of the subtlety in model generation methods.

The advantages are that (i) only forward calculations are made and (ii) some probabilities can be obtained on model parameters. Best for sparsely parameterized models. One needs to be careful that bounds on explored model space don't unduly influence the outcome.